

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO1

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of Artificial Intelligence

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the class of Artificial Intelligence that would surpass human ability at every task.

- A) Artificial Super Intelligence <-- correct
- B) Artificial Narrow Intelligence
- C) Artificial General Intelligence
- D) Artificial Reactive Intelligence

Answer: A. Super intelligence is the hypothetical stage beyond general intelligence. Everything in service today is narrow.

Q2. Name the functionality class of Artificial Intelligence that keeps no memory of earlier inputs.

- A) Limited memory systems
- B) Reactive machines <-- correct
- C) Theory of mind systems
- D) Self-aware systems

Answer: B. A reactive machine responds only to the situation presently in front of it, with no record of what came before.

Q3. State the component of an intelligent agent that receives information from the environment.

- A) Its actuators
- B) Its performance measure
- C) Its sensors <-- correct
- D) Its goal state

Answer: C. Sensors deliver the percepts. What they fail to reveal is exactly what makes an environment partially observable.

Q4. Identify the stage of the Artificial Intelligence development lifecycle in which the problem and its success criteria are agreed.

- A) Data preparation
- B) Model training
- C) Deployment
- D) Problem definition <-- correct

Answer: D. Settling what the system is for, and how success will be measured, before any data is gathered is what keeps the later stages purposeful.

Q5. Name the stage of the lifecycle in which the behaviour of a released system is tracked over time.

- A) Data collection
- B) Monitoring <-- correct
- C) Feature engineering
- D) Model selection

Answer: B. Monitoring detects the decay in performance that follows when the incoming data drifts away from the training distribution.

Q6. Define the agent function.

- A) A mapping from every action to a numeric reward
- B) A mapping from every state to its utility value
- C) A mapping from every sensor to an actuator

D) A mapping from every percept sequence to an action <-- correct

Answer: D. The agent function is the abstract specification of behaviour; the agent program is the concrete implementation that computes it.

Q7. Identify the type of environment in which the outcome of an action is not certain.

- A) Stochastic <-- correct
- B) Deterministic
- C) Episodic
- D) Discrete

Answer: A. In a stochastic environment the same action from the same state can lead to different results, so the agent must be prepared to replan.

Q8. Name the agent type that maintains an internal model of the part of the world it cannot currently see.

- A) A simple reflex agent
- B) A table-driven agent
- C) A model-based reflex agent <-- correct
- D) A utility-based agent

Answer: C. The internal state lets the agent behave sensibly when the present percept alone cannot distinguish two genuinely different situations.

Q9. State the Responsible Artificial Intelligence principle that requires similar cases to be treated alike.

- A) Privacy
- B) Transparency
- C) Fairness <-- correct
- D) Accountability

Answer: C. Fairness concerns systematic differences in outcome between groups that ought to be treated the same way.

Q10. Identify the term for a confidently stated but false claim produced by a generative model.

- A) A hallucination <-- correct
- B) An outlier
- C) A false negative
- D) An overfit

Answer: A. The model is trained to produce fluent continuations rather than to check facts, so a plausible but non-existent statement can be produced with full confidence.

Q11. Name the field that draws on statistics, programming and domain knowledge to obtain insight from data.

- A) Deep learning
- B) Robotics
- C) Computer vision
- D) Data science <-- correct

Answer: D. Data science overlaps with machine learning but takes in collection, cleaning, analysis and the communication of findings as well.

Q12. Identify the manufacturing application of Artificial Intelligence that anticipates when a machine will fail.

- A) Payroll processing
- B) Predictive maintenance <-- correct
- C) Inventory labelling
- D) Document archiving

Answer: B. Sensor histories are used to predict a failure before it happens, so that the machine can be serviced during planned downtime.

Q13. State the term for the instruction text supplied to a generative model.

- A) A percept
- B) A label
- C) A feature
- D) A prompt <-- correct

Answer: D. The prompt shapes one reply and changes nothing about the model itself, which is why prompt design is cheap and reversible.

Q14. Name the healthcare application of Artificial Intelligence that reads scans in support of a specialist.

- A) Appointment scheduling
- B) Payroll calculation
- C) Medical image analysis <-- correct
- D) Stock ordering

Answer: C. Diagnostic imaging is among the most developed healthcare applications, and it supports rather than replaces the clinician's judgement.

Q15. Identify the environment property that holds when several agents are acting at the same time.

- A) Single-agent
- B) Multi-agent <-- correct
- C) Static
- D) Episodic

Answer: B. A multi-agent environment contains other decision makers whose choices affect the outcome, which introduces competition or a need to coordinate.

Q16. Name the smart-city application of Artificial Intelligence that adjusts signal timings from live traffic.

- A) Adaptive traffic control <-- correct
- B) Payroll automation
- C) Document translation
- D) Image compression

Answer: A. Adaptive signalling, along with waste routing and energy management, is among the standard smart-city applications on the syllabus.

Q17. Explain why deep learning is described as a subset of machine learning.

- A) It was developed before machine learning existed
- B) It is one family of methods for learning from data <-- correct
- C) It needs no training data of any kind at all
- D) It applies only to problems written out as rules

Answer: B. Machine learning is any approach that learns behaviour from data; deep learning is the branch that does so with many-layered neural networks.

Q18. Distinguish a goal-based agent from a utility-based agent.

- A) A goal-based agent tests whether the goal is met, a utility-based one grades outcomes <-- correct
- B) A utility-based agent tests whether the goal is met
- C) Both of the two only test whether the goal is met
- D) Both of the two grade every outcome numerically

Answer: A. When two action sequences both reach the goal, only a utility function can say which of them is preferable.

Q19. Describe what it means for a task environment to be static.

- A) It contains exactly one agent at any moment
- B) It has a finite number of distinct states
- C) It gives the agent complete percepts always
- D) It does not change while the agent is deliberating <-- correct

Answer: D. In a static world a plan is still valid when the agent finishes computing it. In a dynamic world the plan may already be out of date.

Q20. Interpret the following: an agent's sensors reveal only the room it currently occupies and nothing of the rest of the building.

- A) The environment is fully observable
- B) The environment is deterministic
- C) The environment is partially observable <-- correct
- D) The environment is episodic

Answer: C. The percepts do not reveal the complete state, so the agent must maintain an internal record of what it has seen elsewhere.

Q21. Predict what happens to a deployed model when the data it meets in service drifts away from its training data.

- A) Its accuracy falls although the code is unchanged <-- correct
- B) Its accuracy improves as it sees more data
- C) It automatically retrains itself on the new data
- D) It refuses to produce any further predictions

Answer: A. The model reproduces the relationship present when it was fitted. Drift is detected by monitoring and corrected by retraining.

Q22. Recognize the setting described: a robot arm learns to grasp an unfamiliar object by trying and receiving a score after each attempt.

- A) Supervised classification of grasps
- B) Unsupervised clustering of objects
- C) Reinforcement learning applied to robotics <-- correct
- D) Rule-based expert reasoning

Answer: C. The signal is a reward rather than the correct action, so the agent has to discover a good policy by acting and observing the score.

Q23. Explain how generative Artificial Intelligence differs from a classifier.

- A) It assigns a label rather than composing content
- B) It composes new content rather than assigning a label <-- correct
- C) It requires far less training data than a classifier
- D) It cannot be applied to text of any kind

Answer: B. A classifier separates existing inputs into categories. A generative model produces text, images or code that did not previously exist.

Q24. Summarize what the Artificial Intelligence development lifecycle covers.

- A) Only the training of the model on the data
- B) Only the writing of the software interface
- C) Only the collection and cleaning of the data
- D) Everything from defining the problem to monitoring the release <-- correct

Answer: D. Training is one stage among several, and the stages after release are as important as those before it.

Q25. Distinguish a deterministic environment from a stochastic one.

- A) In a stochastic one the action fixes the next state
- B) Both of the two have next states fixed by the action
- C) In a deterministic one the action fixes the next state <-- correct
- D) Neither of the two has any defined next state

Answer: C. Chance intervening in the outcome is what makes an environment stochastic, and it is why a robot on a real floor is harder to control than one in simulation.

Q26. Explain why a system that affects people must have an accountable owner.

- A) The software licence requires a named owner
- B) The model runs faster when an owner is named
- C) An owner improves the accuracy of the model
- D) Someone must answer for an outcome the system produces <-- correct

Answer: D. Accountability prevents the system itself being offered as the explanation for a decision that harmed somebody.

Q27. Interpret a system that continues to make accurate predictions but is being used on a population quite unlike the one it was built for.

- A) Its reported accuracy is not evidence for this population <-- correct
- B) Its accuracy will improve on the new population
- C) The population makes no difference to the model
- D) The model will refuse to run on the new data

Answer: A. Performance figures apply to data resembling the evaluation set. Extending a system to a new population requires it to be evaluated again on that population.

Q28. Predict the risk of using a generative model to answer legal questions with no expert check.

- A) It will refuse to answer any legal question
- B) It may state a plausible but non-existent provision <-- correct
- C) It will always cite the wrong section number
- D) It will answer far more slowly than usual

Answer: B. Fluency and correctness are separate properties, and a confident, well-formed answer carries no guarantee that the provision it names exists.

Q29. Describe what makes a task environment continuous rather than discrete.

- A) It contains an unlimited number of agents
- B) It never changes while the agent deliberates
- C) It gives the agent access to the whole state
- D) Its states and actions vary smoothly rather than in steps <-- correct

Answer: D. Steering a vehicle is continuous; a board game is discrete. The distinction decides whether the state can be enumerated at all.

Q30. Explain the difference between data science and machine learning.

- A) Machine learning is broader and includes data science
- B) Data science is broader and aims at insight for people <-- correct
- C) The two terms describe exactly the same activity
- D) Data science never makes use of any data at all

Answer: B. Machine learning is one set of techniques used within data science, which also covers collection, cleaning, analysis and the presentation of findings.

Q31. Distinguish a learning agent from an agent whose rules are fixed.

- A) A fixed agent improves from the outcomes it observes
- B) Both of the two improve from what they observe

C) A learning agent improves from the outcomes it observes <-- correct

D) Neither of the two responds to its current percept

Answer: C. A fixed agent repeats the same mistake indefinitely; a learning agent changes the way it chooses in the light of feedback.

Q32. Explain why Artificial Intelligence in healthcare is normally used to support rather than replace a clinician.

- A) The clinician remains accountable and sees context the model cannot <-- correct
- B) The model is far slower than the clinician is
- C) The model cannot process any medical images
- D) Regulations forbid any use of AI in medicine

Answer: A. The system narrows the field or flags a finding; the judgement, and the responsibility for it, stays with a person who can weigh the whole case.

Q33. Recognize the ethical issue described: a model's training records contain far fewer examples from one section of the population.

- A) Representation bias in the training data <-- correct
- B) Overfitting to the training data
- C) A leak between the training and test data
- D) An error in the choice of the loss function

Answer: A. The model has less evidence about that group and usually performs worse for it, which is why results must be reported broken down by group.

Q34. Summarize the applications of Artificial Intelligence in robotics named on the syllabus.

- A) Payroll processing and document archiving
- B) File compression and network routing
- C) Spreadsheet calculation and report printing
- D) Perception, navigation and manipulation by autonomous robots <-- correct

Answer: D. A robot must perceive its surroundings, decide where to go and act physically upon the world, which is the full agent cycle in a physical setting.

Q35. Explain why the way a project will be evaluated should be settled at the start.

- A) It reduces the cost of the computing hardware
- B) It decides what is built and what counts as success <-- correct
- C) It removes the need to collect any data
- D) It guarantees the model will be accurate

Answer: B. Without an agreed measure the project cannot be shown to have succeeded, and disagreements about its purpose surface only when it is finished.

Q36. Interpret a system described as reactive in the functionality classification.

- A) It remembers everything it has ever been shown
- B) It reasons about the beliefs of other agents

C) It responds to the present input and remembers nothing <-- correct

D) It holds an internal model of its own state

Answer: C. A reactive system maps the current situation straight to a response, which is adequate for a thermostat but not for anything requiring context.

Q37. Predict the effect of deploying a model with no plan for monitoring it.

A) The model will stop running after a fixed period

B) The model will improve steadily without help

C) A gradual decline in performance will go unnoticed

<-- correct

D) The model will retrain itself when it drifts

Answer: C. Nothing about the model announces that it has stopped working. Without measurement, decisions continue to be made on a system that no longer performs.

Q38. Describe what the privacy principle requires of a system that uses personal records.

A) That the data is published for public inspection

B) That only necessary data is collected and kept securely <-- correct

C) That every record is copied to a second server

D) That the model is retrained every single week

Answer: B. Privacy covers what is collected, how long it is retained, who may see it and whether it ever leaves the organisation.

Q39. Explain why generative Artificial Intelligence raises questions about authorship and consent.

A) It cannot produce any text longer than a page

B) It stores a copy of every question it is asked

C) It requires a licence for every generated word

D) It is trained on work produced by identifiable people <-- correct

Answer: D. The training corpus contains material whose creators did not necessarily agree to that use, and the output can resemble their work closely.

Q40. Distinguish an agent's environment from its architecture.

A) The environment is the world, the architecture the machinery <-- correct

B) The architecture is the world, the environment the machinery

C) Both of the two describe the surroundings of the agent

D) Both of the two describe the hardware of the agent

Answer: A. The architecture is the computing platform with its sensors and actuators; the environment is what it perceives and acts upon.

----- End of Sample Question Set -----